

SAISHRUTI SAIRAM VEDHA

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SUMMARY

M.S. Electrical Engineering candidate at NYU focused on ASIC design, verification, and physical design. Wrote synthesizable SystemVerilog RTL for an FSM-controlled MBIST block, closed timing through synthesis, STA, and SDC checks, and completed DRC/LVS, parasitic extraction, and PVT analysis on custom SRAM and flip-flop layouts. Built C++ RISC-V simulators validated on 50+ scenarios, and automated regression and debug flows in Python, Tcl, and Linux.

EDUCATION

New York University, Tandon School of Engineering — M.S. Electrical Engineering | Expected May 2027

SRM Institute of Science and Technology, India — B.Tech Electronics & Communication Engineering | May 2025

TECHNICAL SKILLS

Digital Design & Verification: SystemVerilog, Verilog, RTL Design, FSM Design, Simulation-Based Verification, Waveform Debug, SRAM/MBIST, FPGA (Vivado), ModelSim, Verilator, Synopsys VCS

Physical Design & Timing: Static Timing Analysis, Setup/Hold Analysis, SDC Constraints, Timing Closure, PVT Analysis, DRC/LVS, Parasitic Extraction, Post-Layout Verification, Signal Integrity, PPA Analysis

Tools & Automation: Synopsys Design Compiler, Cadence Virtuoso, HSPICE, Python, Tcl, Shell, C++, C, Linux/Unix, Git

Architecture & Hardware: RISC-V, 5-Stage Pipeline, Hazard Handling, Memory Subsystems, STM32, UART, SPI, I2C, Oscilloscope, Logic Analyzer, KiCad

ASIC & ARCHITECTURE PROJECTS

MBIST ASIC Implementation & Timing Closure, 64×8 SRAM | SystemVerilog, Design Compiler, SDC, VCS

- Designed synthesizable RTL for an FSM-controlled MBIST subsystem with SRAM interfaces, comparator/decoder blocks, and multiplexed datapaths.
- Ran synthesis, STA, and setup/hold analysis to find critical paths and verify timing; refined RTL and constraints to resolve functional and timing violations across memory-test scenarios.

6T SRAM Physical Design & PVT Analysis | Cadence Virtuoso, HSPICE

- Built a transistor-level 6T SRAM schematic and custom layout; completed DRC/LVS and parasitic extraction.
- Characterized access timing, read/write stability, and static noise margin across process, voltage, temperature, and extracted-parasitic conditions.

D Flip-Flop Full-Custom Layout and 32kB SRAM Wordline RC Analysis | Cadence Virtuoso, SPICE

- Quantified layout-dependent delay of a full-custom D flip-flop through DRC/LVS, extraction, and post-layout timing.
- Modeled wordline RC in a 32kB SRAM array to isolate wire resistance, capacitance, and loading as sources of access-latency degradation.

32-bit RISC-V Simulator and 5-Stage Pipeline Model | C++, Python, Linux

- Implemented decode, ALU, register file, memory access, and branching in modular C++; validated across 50+ instruction-level scenarios using register, memory, and execution-trace analysis.
- Modeled hazard detection, forwarding, branch handling, and stalls in a 5-stage pipeline.

AI-Enabled Embedded Monitoring & Alert System (Animal Intrusion Detection) | Python, YOLO, Embedded Hardware | Best Paper, IEEE RAEEUCCI 2025

- Built an end-to-end edge system that runs real-time YOLO object detection on embedded camera hardware and triggers automated alerts on detection.
- Integrated sensing, processing, and communication components, and validated the full pipeline through system-level testing and field deployment.

HeatStop.AI — AI Agent Platform

- End-to-end AI platform processing 50K+ records from 8 NYC data sources: Python ETL pipelines, PostgreSQL storage with query optimization, Redis caching delivering sub-100ms queries, Claude API + LangChain RAG with 4 deterministic guardrails reducing hallucination rate from 12% to <1%; FastAPI backend with OAuth2/JWT/RBAC; deployed on AWS (S3, CloudWatch, IAM).
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RESEARCH & EXPERIENCE

NYU VIP Team — Processor Hardware & RTL Validation Research | Ongoing

- Using Verilator, Python, and Linux to simulate RISC-V processor RTL, trace execution, and debug at the architecture level.
- Built automated workflows for regression testing, workload analysis, and root-cause analysis.

New York University — Course Assistant, Digital Logic & FSM Design | Jan 2026 – Present

- Guided 120+ students through RTL implementation, FPGA validation, and waveform debugging with ModelSim and Vivado; diagnosed timing violations and reset failures from simulation and timing reports.

e-con Systems — Embedded Software Engineering Intern | Summer 2026

- Re-engineered legacy C++/C# camera tools into Python utilities that send commands to live embedded cameras over Ethernet and read back each device's MAC ID and OTP number in real time, automating device configuration, programming, validation, and testing; root-caused failures across the hardware/software boundary.

Shibaura Machine India Pvt. Ltd. — Automation Engineering Intern | Aug 2024 – Oct 2024

- Built Python automation and data-analysis utilities for real-time monitoring, testing, and hardware/software debugging.

Torus Robotics Pvt. Ltd. — Embedded Systems Intern | Dec 2023

- Supported PCB design and board bring-up for STM32 systems; debugged signal-level issues with oscilloscopes and logic analyzers.
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ACHIEVEMENTS

Best Paper Award, IEEE RAEEUCCI 2025 | 1st Place, HACKFEST 2 & Poster Expo 25 | Scopus-indexed research, AISC 2024 | Chancellor Award for Academic Excellence